Analysis and cooking status

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Pass1 status

workflow_name	jobs	success
rgd-dmra-pass1_CxC_2-18339x116	36705	6752 (18.40%)
rgd-dmra-pass1_CuSn_1-18561x33	14687	14597 (99.39%)
rgd-dmra-pass1_CuSn_2-18347x294	70791	5000 (7.06%)
rgd-dmra-pass1_empty_nominal-18399x2	113	111 (98.23%)
rgd-dmra-pass1_empty_zero_field-18316	239	238 (99.58%)
rgd-dmra-pass1_LD2_1-18431x16	7702	7702 (100.00%)
rgd-dmra-pass1_CxC_1-18445x13	6873	6873 (100.00%)
rgd-dmra-pass1_LD2_2-18305x144	40076	0 (0.00%)

- Path: `/cache/clas12/rg-d/production/pass1/recon`
- Total: 177186, Done: 41273, Running: 15063, Pending: 120850
- (Done/Total): 23.29%, (Running/Total): 8.50%, (Pending/Total): 68.21%

Pass1 status

workflow_name	jobs	success
rgd-dmra-pass1_CxC_2-18339x116	36705	6752 (18.40%)
rgd-dmra-pass1_CuSn_1-18561x33	[illegible]	[illegible]
rgd-dmra-pass1_CuSn_2-18347x294	[illegible]	[illegible]
rgd-dmra-pass1_empty_nominal-18399x	[illegible]	[illegible]
rgd-dmra-pass1_empty_zero_field-1831	[illegible]	[illegible]
rgd-dmra-pass1_LD2_1-18431x16	[illegible]	[illegible]
rgd-dmra-pass1_CxC_1-18445x13	[illegible]	[illegible]
rgd-dmra-pass1_LD2_2-18305x144	40076	0 (0.00%)

```
--------------- rgd-dmra-pass1_CuSn_1-18561x33 ---------------
     total:     14597 /     14687 = 99.3872%
    decmrg:      9524 /      9524 = 100.0000%
     recon:      4766 /      4770 = 99.9161%
       ana:       291 /       327 = 88.9908%
    anamrg:        13 /        33 = 39.3939%
  anaclean:         3 /        33 = 9.0909%
```

- Path: `/cache/clas12/rg-d/production/pass1/recon`
- Total: 177186, Done: 41273, Running: 15063, Pending: 120850
- (Done/Total): 23.29%, (Running/Total): 8.50%, (Pending/Total): 68.21%

clas12-config: Simulation and reconstruction

- Goal: Add the GCARDS and YAML files to clas12-config to be able to use OSG with RG-D:
  - For that, some modification in the detector geometry is needed
  - We need 1 variation per target named rgd_fall2023_X
- I have changed the geometry.pl script and the associated GXML file to be able to do that.

CxC variation:

```
<!-- RG-D Nuclear-target Assembly; variation can be for the flag assembly: lD2CuSn, lD2CxC -->
<detector name="/work/clas12/ouillon/tmp/ct2/experiments/clas12/targets/target" factory="TEXT" variation="rgd_fall2023_CxC"/>
<detector name="/work/clas12/ouillon/tmp/ct2/experiments/clas12/targets/cad/" factory="CAD" variation="rgd_fall2023_CxC"/>
```

LD2 variation:

```
<detector name="/work/clas12/ouillon/tmp/ct2/experiments/clas12/targets/target" factory="TEXT" variation="rgd_fall2023_lD2"/>
<detector name="/work/clas12/ouillon/tmp/ct2/experiments/clas12/targets/cad/" factory="CAD" variation="rgd_fall2023_lD2"/>
```

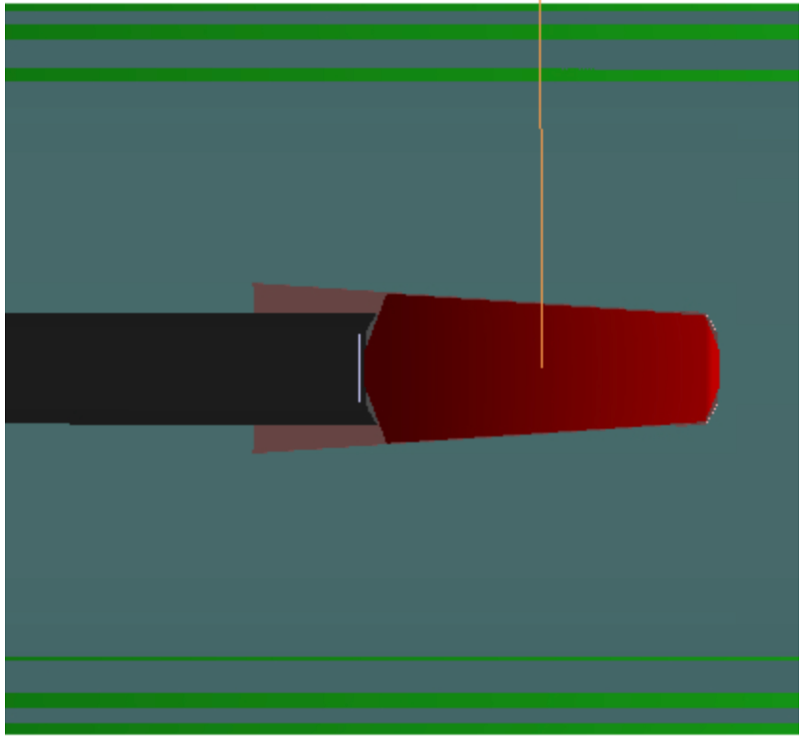
CuSn variation:

```
<detector name="/work/clas12/ouillon/tmp/ct2/experiments/clas12/targets/target" factory="TEXT" variation="rgd_fall2023_CuSn"/>
<detector name="/work/clas12/ouillon/tmp/ct2/experiments/clas12/targets/cad/" factory="CAD" variation="rgd_fall2023_CuSn"/>
```

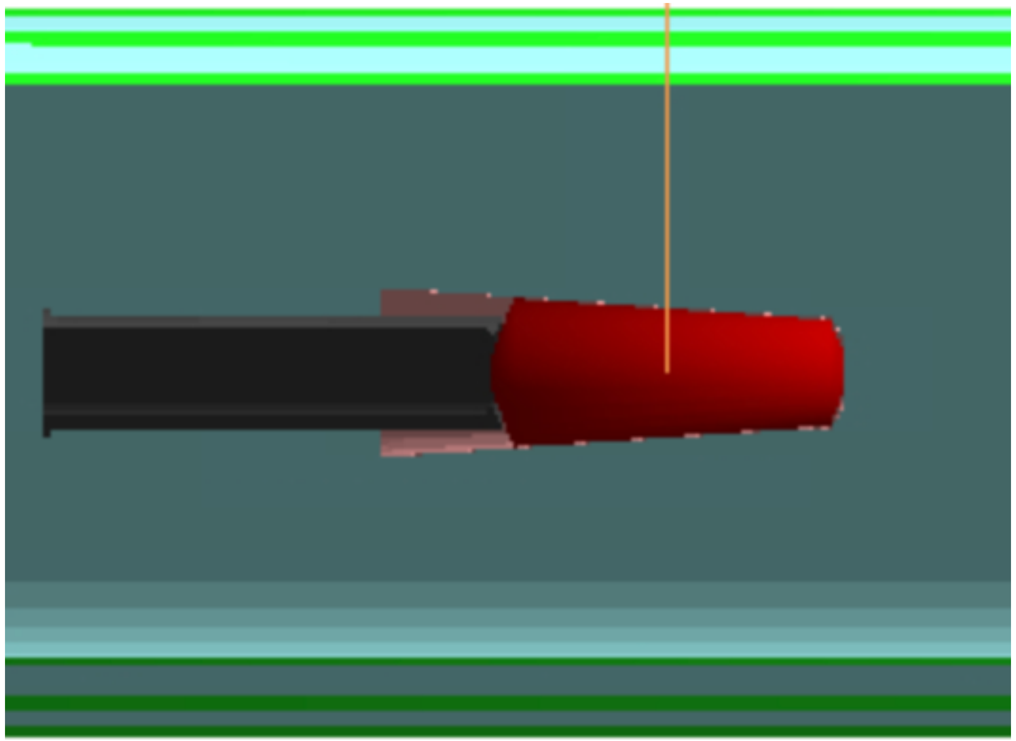
Work on GEMC/GCARD: Result

- Orange line is at -5.5cm

New geometry: LD2



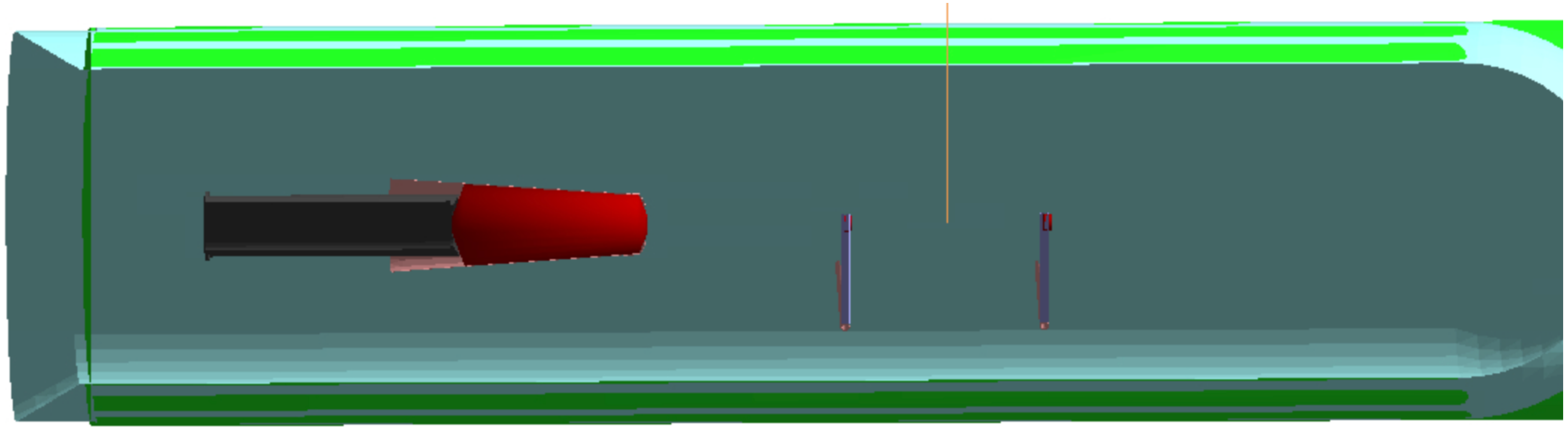
Old geometry: LD2



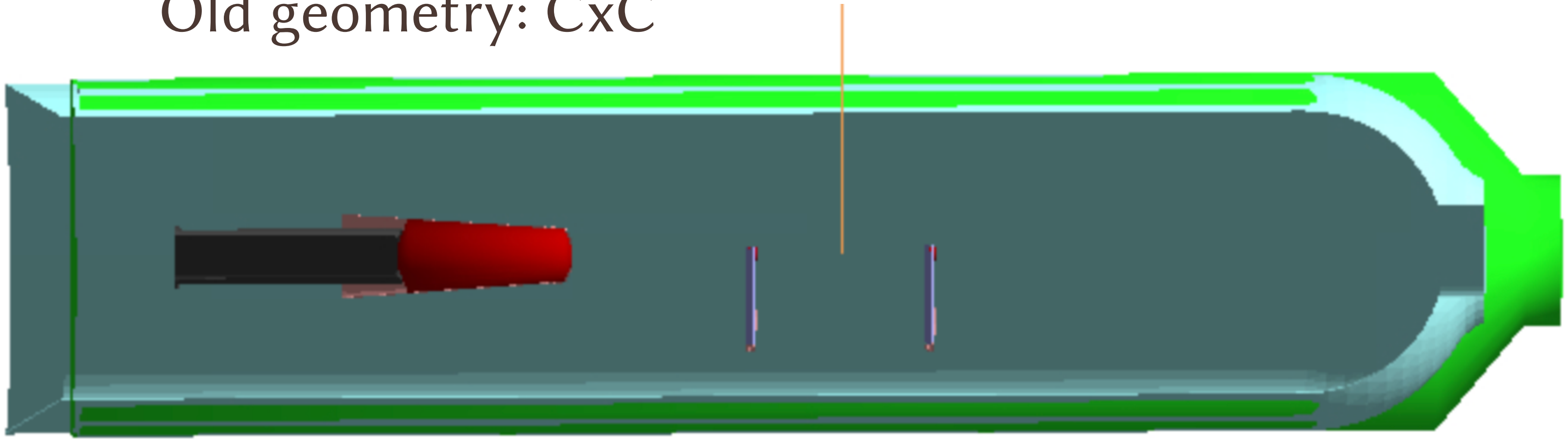
Work on GEMC/GCARD: Result

- Orange line is at -5.5cm

New geometry: CxC



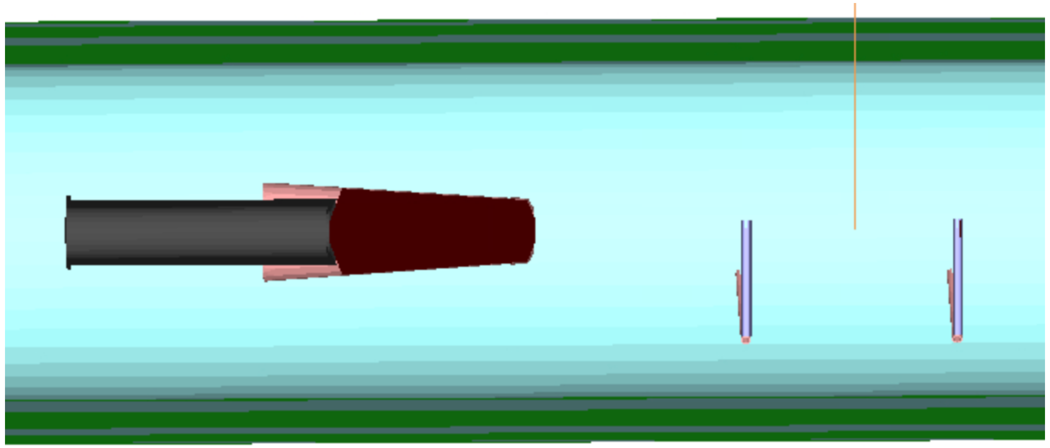
Old geometry: CxC



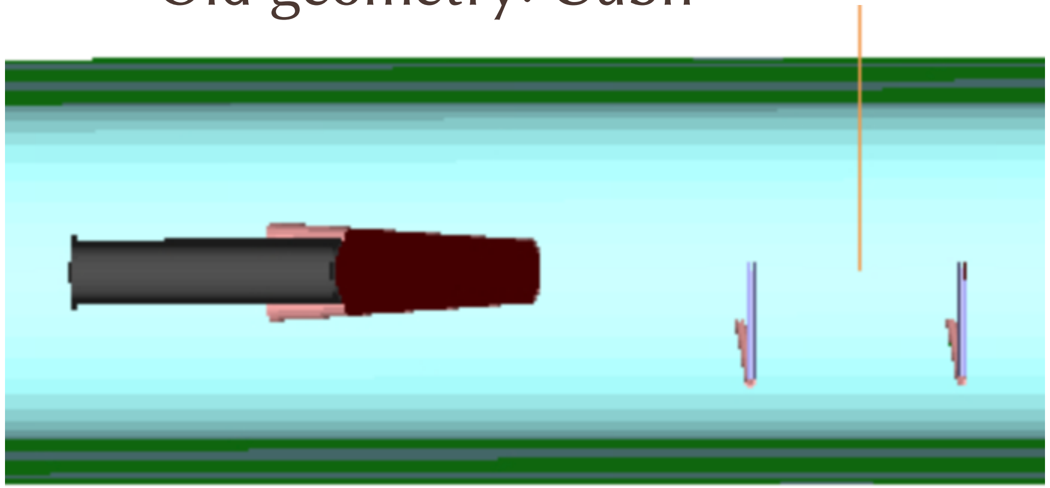
Work on GEMC/GCARD: Result

- Orange line is at -5.5cm

New geometry: CuSn



Old geometry: CuSn

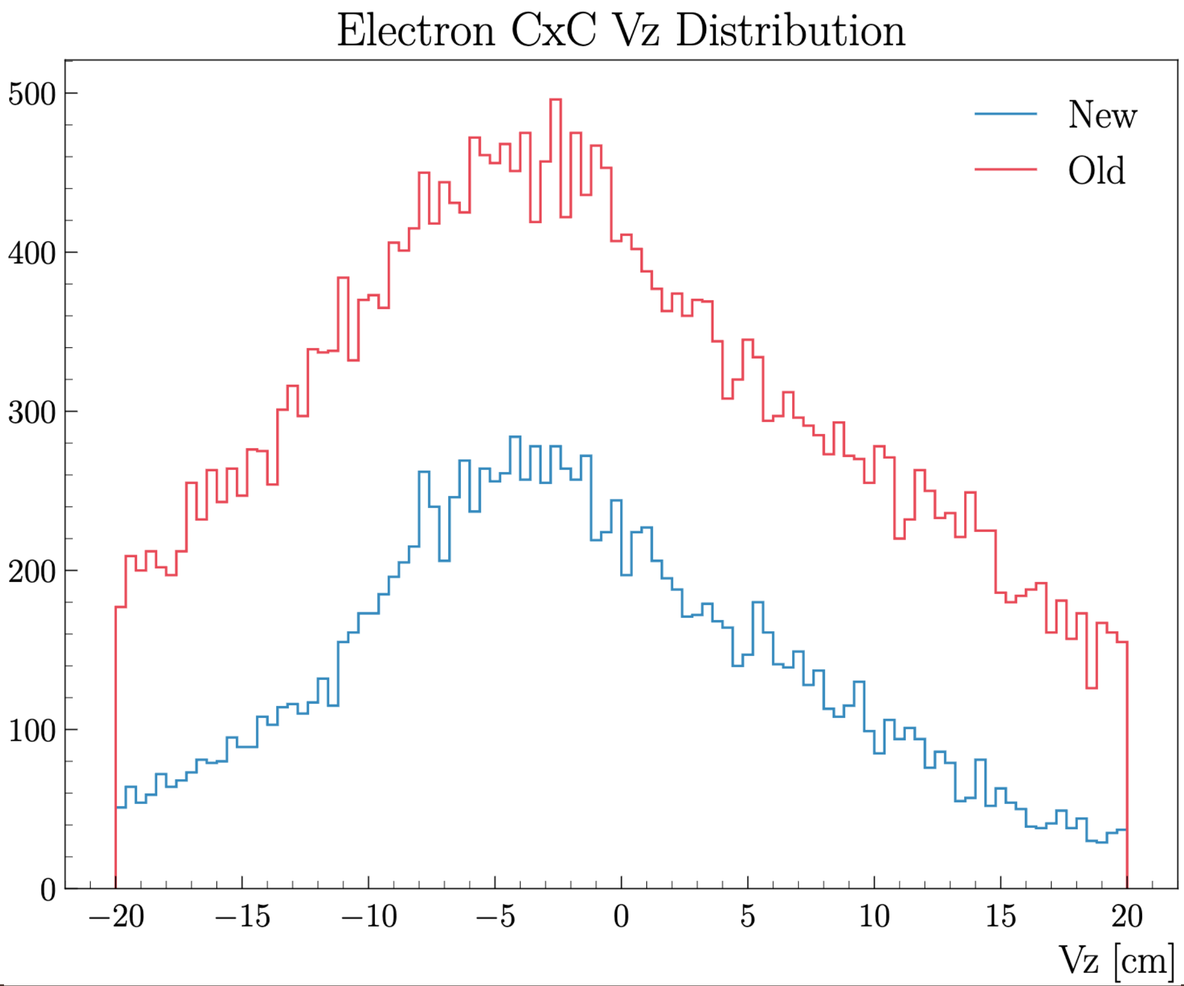


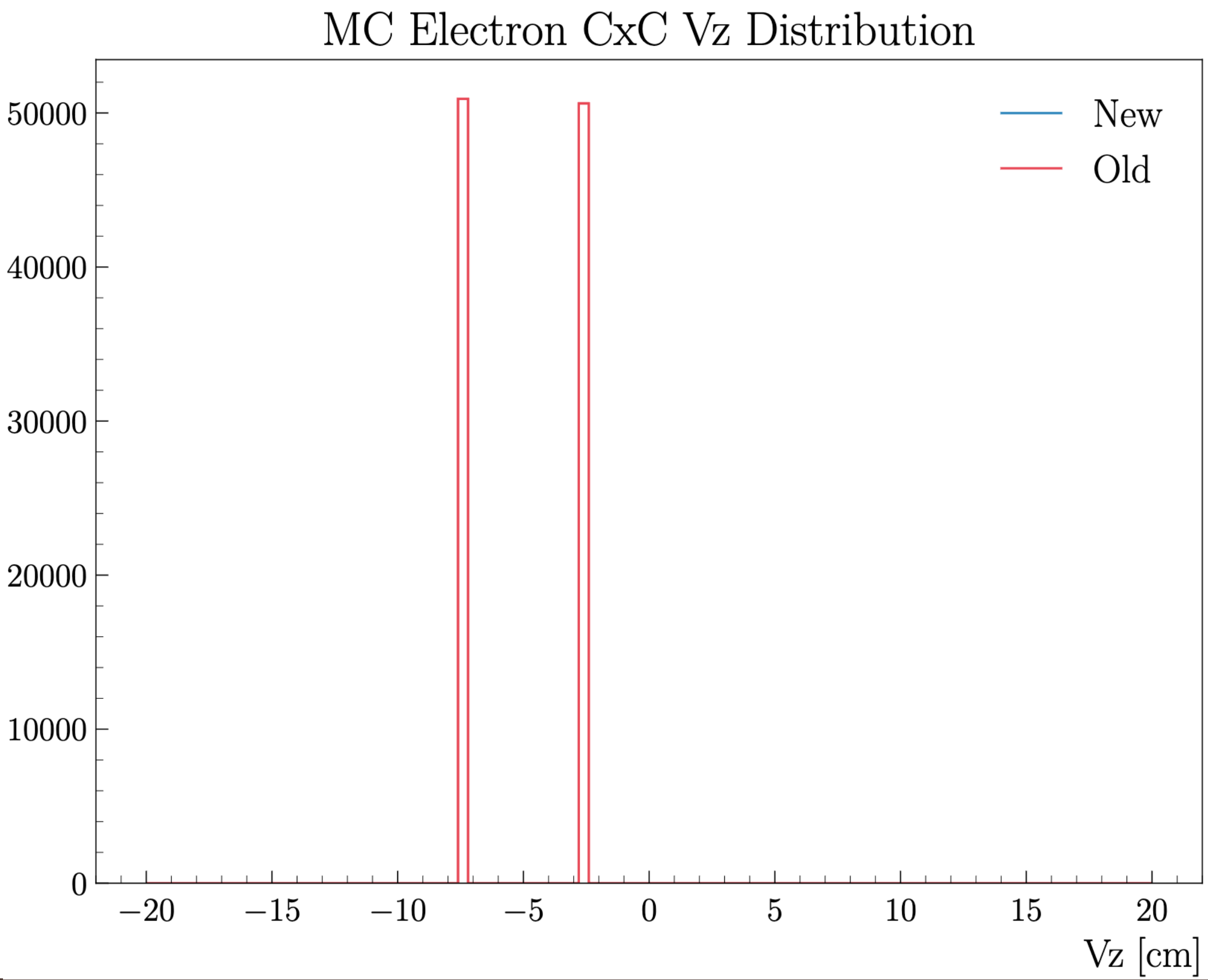
Work on GEMC/GCARD: Result

Simulation GEMC/dev for new, 5.11 for Old

COATJAVA 13.0.0

Problem





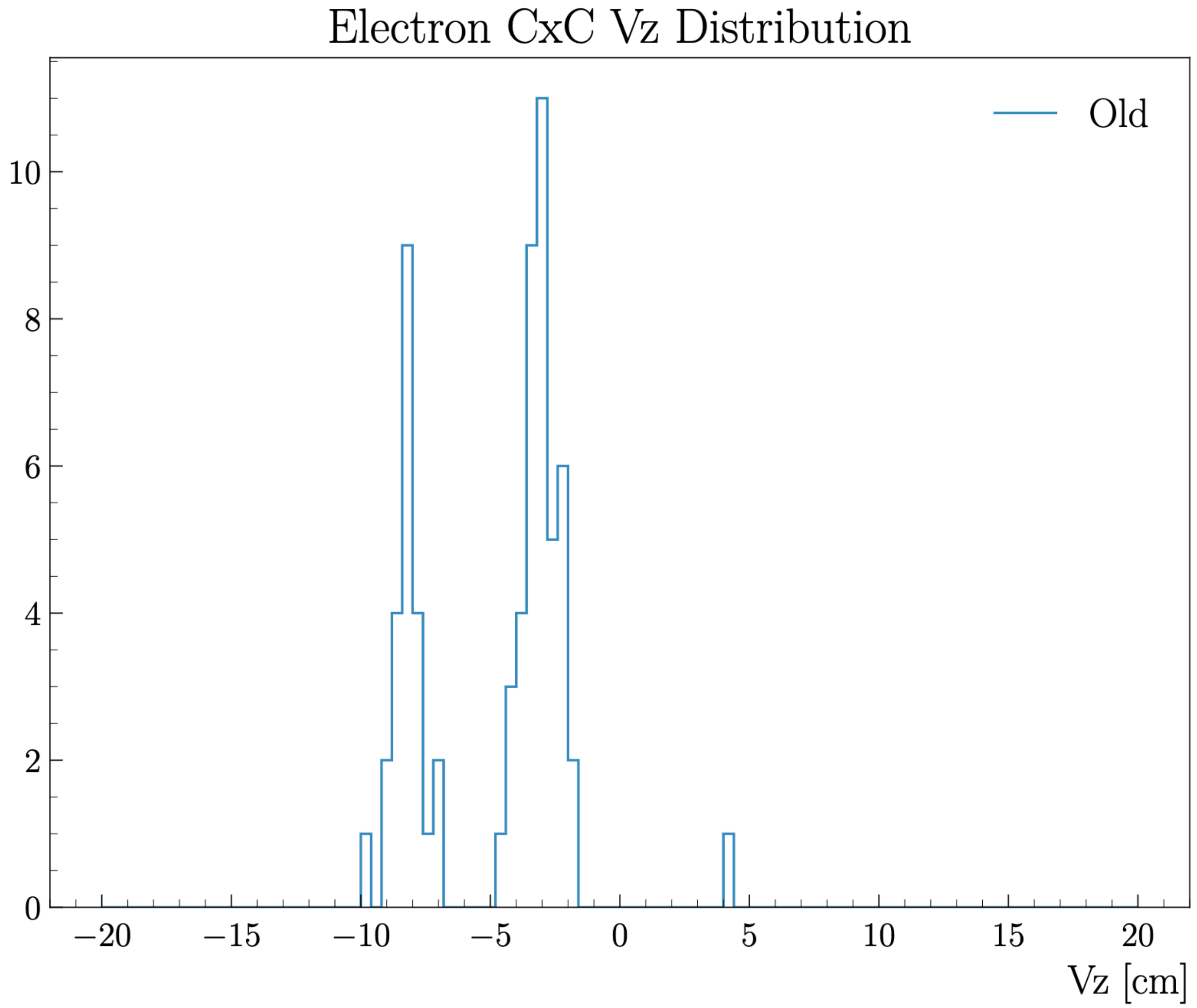
Work on GEMC/GCARD: Result

GEMC 5.11 / COATJAVA 11.1.0

The problem arises from the DC reconstruction and simulation.

The time-to-distance functional form has been changed in the reconstruction, but not in the simulation.

Therefore, we need to use a version of COATJAVA where the t2d function matches the one used in the simulation. This means we should use version 11.1.0 or earlier.



- Variation:
  - Old CuSn shifted by -15.5 cm

- So, with the previous shift of -15 cm, the first foil of the target was at -7.5 cm.
- But with a shift of -15.5 cm the target is at -8 cm. So, the simulation (Lund files) needs to be changed.